INTERNATIONAL MONETARY FUND

MYANMAR

June 16, 2020

REQUESTS FOR DISBURSEMENT UNDER THE RAPID CREDIT FACILITY AND PURCHASE UNDER THE RAPID FINANCING INSTRUMENT—DEBT SUSTAINABILITY ANALYSIS[1]

Approved By
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Prepared by the staffs of the International Monetary Fund and the International Development Association

Risk of external debt distress:	Low[2]
Overall risk of debt distress:	Low
Granularity in the risk rating	Not applicable
Application of judgement:	No
Macroeconomic projections	The main changes in the current macroeconomic assumptions, compared with those in the 2019 DSA, stem from the estimated impact of COVID-19 (see text table for detailed comparison). Assuming a localized outbreak, real GDP growth is expected to slow down considerably to 1.4 percent in FY19/20. The current account deficit is expected to widen due to lower garments and gas exports, higher medical imports, and weaker tourism and remittances. FDI is expected to slow as projects are delayed. Inflation is expected to decline reflecting commodity prices and subdued domestic demand. The fiscal stimulus required to manage the impact of the pandemic together with weak revenues will widen the budget deficit by about 2 percent of GDP annually over the next two years. A gradual recovery starting Q1 FY20/21 is projected.[3]
Financing strategy	To keep adequate reserves given uncertainties around the length of the pandemic and preexisting financial sector vulnerabilities, an external financing gap amounting to US$1.7 billion is expected in FY19/20. The external gap, which is expected to fill the fiscal gap, will be filled by a mix of external financing (IMF, the World Bank, ADB, EU and Japan, and the DSSI). Such a strategy will help reduce pressure on monetizing the deficit and risks of disorderly external market conditions.

[1] This joint World Bank/IMF DSA has been prepared in the context of the 2020 request for emergency financing from the IMF (RCF/RFI). The macro framework underlying this DSA is the same as that included in the staff report of the 2020 RCF/RFI request which reflects recent global developments. The current macroeconomic framework reflects currently available information. However, updates with respect to economic impact and policy response to the COVID-19 crisis are rapidly evolving and risks are tilted to the downside. Public debt covers the consolidated public sector debt, central bank debt borrowed on behalf of the government, government-guaranteed debt and social security funds. SOE debt is on lent and is therefore included in the coverage of public external debt.

[2] The LIC DSF determines the debt sustainability thresholds by calculating a composite indicator (CI). Based on the CI score (2.72), for Myanmar, the final debt carrying capacity classification for this DSA is medium. This is based on the October 2019 WEO and CPIA 2018.

[3] Myanmar's fiscal year is on October-September basis. All tables and figures are on a fiscal year basis. In the DSA standard tables, for example, 2019 refers to FY2018/19 and ends in September 2019.

Myanmar's risk of external and overall debt distress continues to be assessed as low. External debt indicators are projected to continue on a downward trend; however, rapidly evolving circumstances have raised the projected FY 2019/20 deficit and public debt levels, and made prominent several vulnerabilities (see text table). The IMF's RCF/RFI financing arrangement and the Debt Service Suspension Initiative (DSSI), supported by the G-20 and the Paris Club, will enable a quick increase in priority spending in response to the effects of the pandemic. The DSA shows that, under the baseline, which reflects the COVID-19 shock, all external public and publicly guaranteed (PPG) debt indicators remain below their policy relevant thresholds and benchmarks. A slowdown in exports and the aftermath from a natural disaster are shocks that worsen the debt outlook the most.[4]

Under the baseline, the magnitude of a shock from contingent liability through the financial sector is assumed to be the default 5 percent of GDP. A more prolonged global outbreak could result in more adverse economic outcomes that interact with banking sector fragilities. This could potentially result in the realization of contingent liabilities arising from recapitalization needs. An adverse scenario, which considers the macroeconomic impact from such an assumption, reflects the impact of this downside risk. Here, the PV of public debt-to-GDP ratio deteriorates significantly. It also raises the PV of public debt-to-GDP ratio under the most extreme shock, and breach its benchmark (55 percent) in the medium term suggesting a possible worsening of the risk rating from low to moderate (see figure).

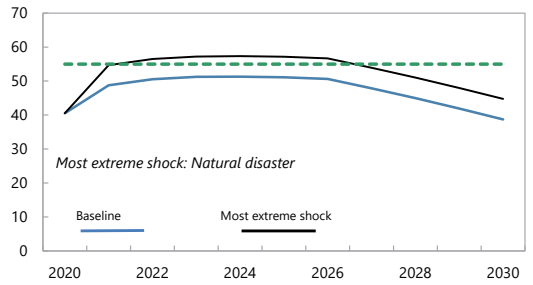
It is imperative that the authorities address the immediate gross financing needs arising from the impact of the COVID-19 crisis, without resorting to potentially more destabilizing central bank financing given Myanmar's experience. At the same time, there may be limited potential to borrowing from domestic banks given vulnerabilities and the risks of crowding out needed credit. Thus, external financing on concessional terms, and the DSSI, will be key to support the policy stimulus while containing risks to price and external stability.[5] *Over time, to avoid an excessive recourse to central bank financing, the authorities should embark on a medium-term revenue strategy underpinned by a revenue target and comprehensive tax policy reforms building on recent administrative reforms. The authorities should remain vigilant of the potential impact from the fragilities in the banking system and put in place a framework for better monitoring and managing fiscal risks including from PPPs. Strengthening debt management capacity remains priority.*

Key Macroeconomic Assumptions, FY2020-24 (Average)

	Baseline	Previous DSA
Real GDP growth (in percent)	5.2	6.3
Inflation (percent change, y/y)	6.2	6.7
Primary fiscal balance (in percent of GDP)	-3.4	-2.5
Overall fiscal balance (in percent of GDP)	-5.3	-4.4
Current account (in percent of GDP)	-4.1	-3.9
FDI (in percent of GDP)	3.4	3.5

Source: IMF staff estimates.

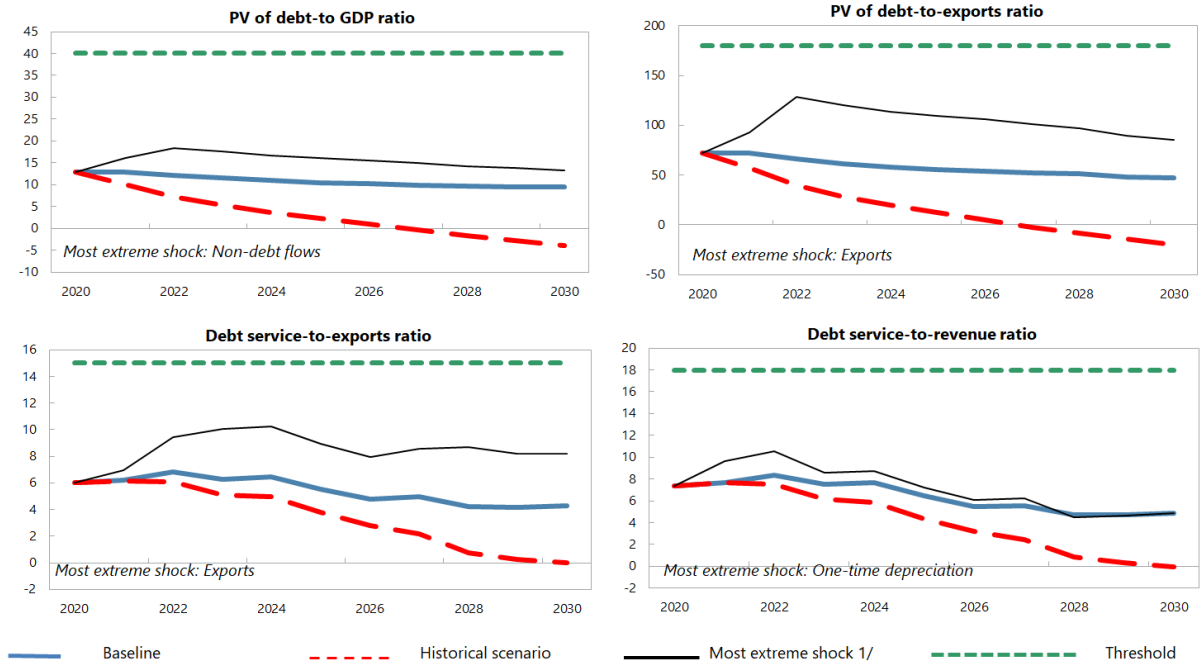
PV of PPG Debt-to-GDP Ratio (Adverse Scenario)



[4] Public/Private investment rate charts are not available in the current DSA from data limitations. Technical assistance from the IMF and various partners continue is ongoing to strengthen macroeconomic data.

[5] Based on data provided on mission, the DSSI is estimated to suspend US$322 million and US$67 million in external debt service in FY2019/20 and FY2020/21 respectively. It would contribute 19 percent of the financing gap in FY2019/20. The savings from suspended debt service in FY2019/20 and FY2020/21 under the DSSI and the related debt service considering this suspension have been incorporated into the macro framework and the DSA.

Figure 1. Myanmar: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2020–2030 1/ 2/



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	Yes	No
Commodity price	n.a.	n.a.
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.0%	5.0%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	25	20
Avg. grace period	6	5

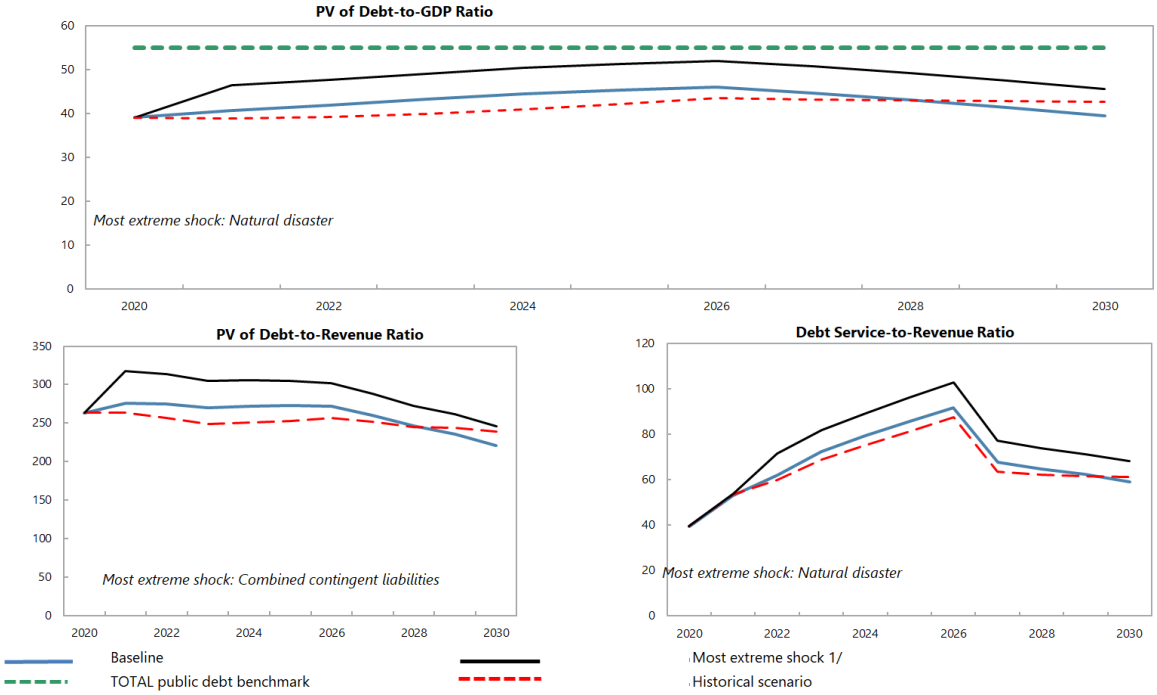
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Myanmar: Indicators of Public Debt Under Alternative Scenarios, 2020–2030 1/



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	14%	14%
Domestic medium and long-term	67%	67%
Domestic short-term	19%	19%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.0%	5.0%
Avg. maturity (incl. grace period)	25	20
Avg. grace period	6	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	1.2%	3.0%
Avg. maturity (incl. grace period)	5	5
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	0.3%	0.3%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

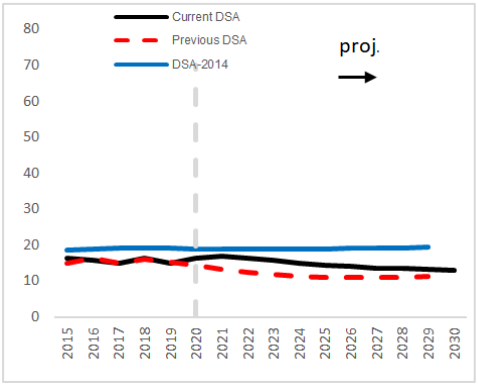
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

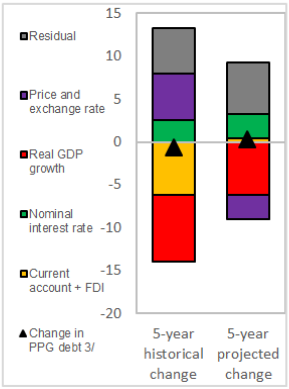
Figure 3. Myanmar: Drivers of Debt Dynamics—Baseline Scenario

External Debt

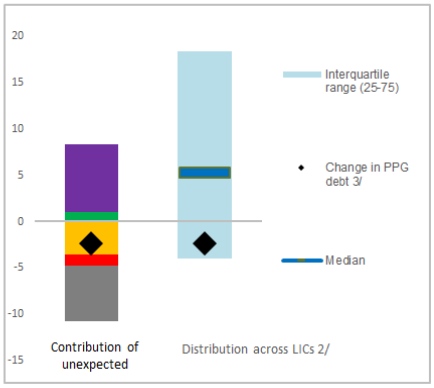
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

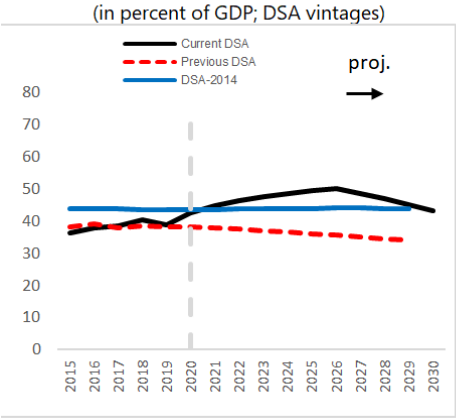


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

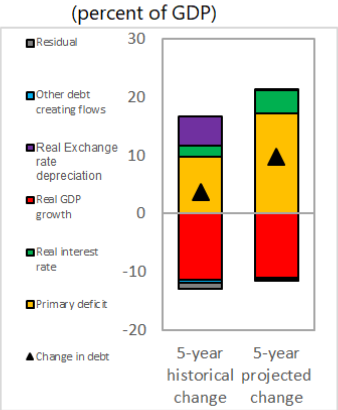


Public Debt

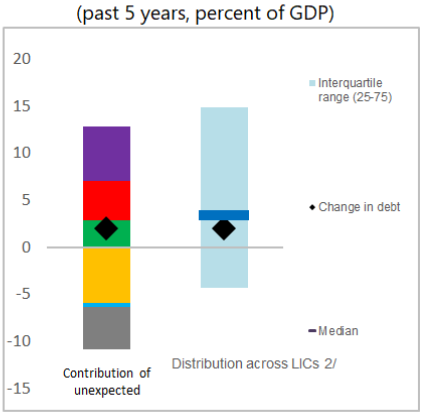
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



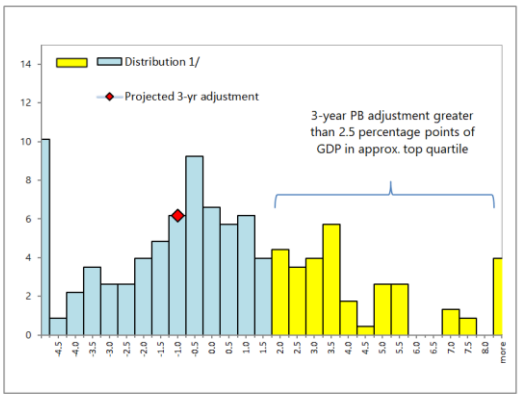
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

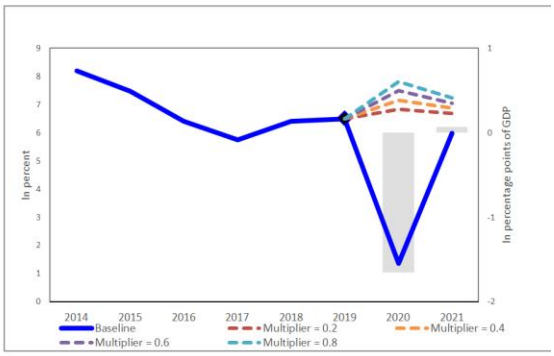
Figure 4. Myanmar: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



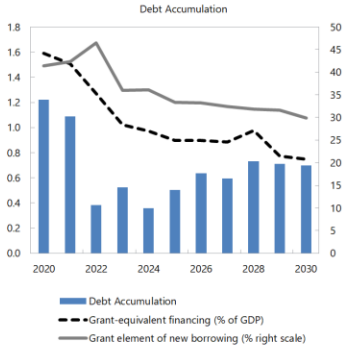
1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

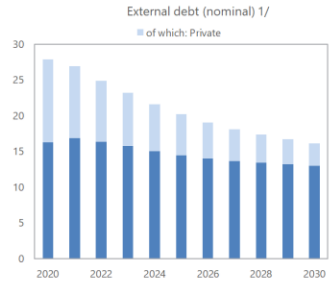
Table 1. Myanmar: External Debt Sustainability Framework, Baseline Scenario, 2017–40

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
External debt (nominal) 1/	**28.9**	**30.9**	**27.7**	**27.9**	**26.9**	**24.9**	**23.2**	**21.6**	**20.2**	**16.1**	**12.7**	**19.3**	**21.1**
of which: public and publicly guaranteed (PPG)	**15.0**	**16.3**	**14.8**	**16.2**	**16.9**	**16.3**	**15.7**	**15.0**	**14.4**	**12.9**	**12.7**	**13.6**	**14.7**
Change in external debt	8.0	2.0	-3.2	0.2	-1.0	-2.0	-1.7	-1.6	-1.4	-0.6	-1.3		
Identified net debt-creating flows	**0.6**	**-2.6**	**-1.1**	**0.9**	**-0.2**	**-0.8**	**-1.2**	**-1.6**	**-1.7**	**-0.8**	**-0.6**	**-2.4**	**-0.9**
Non-interest current account deficit	**6.3**	**4.1**	**1.9**	**3.3**	**3.6**	**3.6**	**3.5**	**3.6**	**3.7**	**4.1**	**4.1**	**2.1**	**3.7**
Deficit in balance of goods and services	7.5	5.2	3.2	3.3	4.1	4.0	4.1	4.1	4.2	4.6	4.9	**1.3**	**4.2**
Exports	21.5	23.3	22.0	17.7	17.7	18.2	18.7	18.8	18.8	19.9	22.4		
Imports	29.0	28.5	25.2	20.9	21.8	22.2	22.8	22.8	23.0	24.6	27.3		
Net current transfers (negative = inflow)	-3.5	-3.4	-3.6	-2.4	-2.7	-2.6	-2.5	-2.4	-2.3	-1.9	-1.5	**-2.1**	**-2.3**
of which: official	-0.4	-0.2	-0.6	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.3	-0.3		
Other current account flows (negative = net inflow)	2.2	2.3	2.3	2.4	2.2	2.1	2.0	2.0	1.8	1.4	0.8	**2.9**	**1.8**
Net FDI (negative = inflow)	**-5.8**	**-4.8**	**-3.1**	**-2.7**	**-2.9**	**-3.4**	**-3.8**	**-4.4**	**-4.5**	**-4.3**	**-4.3**	**-4.1**	**-3.9**
Endogenous debt dynamics 2/	**0.1**	**-1.8**	**0.1**	**0.3**	**-0.9**	**-0.9**	**-0.9**	**-0.9**	**-0.9**	**-0.7**	**-0.5**		
Contribution from nominal interest rate	0.5	0.6	0.7	0.6	0.6	0.6	0.5	0.5	0.4	0.3	0.2		
Contribution from real GDP growth	-1.2	-1.7	-2.0	-0.4	-1.5	-1.5	-1.4	-1.4	-1.3	-1.0	-0.7		
Contribution from price and exchange rate changes	0.8	-0.7	1.3	...	...	...	...	...	...	...	...		
Residual 3/	**7.4**	**4.6**	**-2.1**	**-0.8**	**-0.8**	**-1.2**	**-0.5**	**0.1**	**0.3**	**0.3**	**-0.7**	**3.8**	**-0.1**
of which: exceptional financing	0.0	0.0	0.0	-0.5	-0.1	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**11.8**	**12.8**	**12.8**	**12.1**	**11.5**	**10.9**	**10.4**	**9.4**	**9.4**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**53.7**	**72.4**	**72.3**	**66.2**	**61.6**	**58.1**	**55.6**	**47.1**	**41.7**		
PPG debt service-to-exports ratio	**4.5**	**5.2**	**4.2**	**6.0**	**6.2**	**6.8**	**6.3**	**6.5**	**5.5**	**4.3**	**3.4**		
PPG debt service-to-revenue ratio	**5.5**	**7.0**	**5.9**	**7.4**	**7.7**	**8.4**	**7.6**	**7.7**	**6.4**	**4.9**	**3.9**		
Gross external financing need (Million of U.S. dollars)	4559.5	3766.4	3470.5	4887.8	5057.5	4715.7	4139.7	3697.5	3478.3	4136.5	4807.8		
Key macroeconomic assumptions													
Real GDP growth (in percent)	5.8	6.4	6.5	1.4	6.0	6.2	6.3	6.4	6.5	6.4	5.5	**6.6**	**5.9**
GDP deflator in US dollar terms (change in percent)	-3.7	2.6	-4.2	0.7	2.4	2.9	2.6	2.6	2.6	2.4	2.2	**2.3**	**2.4**
Effective interest rate (percent) 4/	2.4	2.4	2.3	2.3	2.5	2.5	2.3	2.3	2.2	2.1	1.9	**4.2**	**2.3**
Growth of exports of G&S (US dollar terms, in percent)	2.4	18.5	-3.6	-18.1	8.6	12.5	12.1	9.4	9.3	9.8	8.4	**7.6**	**7.8**
Growth of imports of G&S (US dollar terms, in percent)	14.7	7.2	-9.5	-15.3	12.7	11.5	11.8	9.4	10.0	10.0	8.9	**13.4**	**8.4**
Grant element of new public sector borrowing (in percent)	...	...	...	41.5	42.3	46.5	36.0	36.1	33.3	29.9	29.6	**...**	**35.9**
Government revenues (excluding grants, in percent of GDP)	17.5	17.3	15.6	14.4	14.3	14.8	15.6	15.9	16.2	17.5	19.6	**16.8**	**16.0**
Aid flows (in Million of US dollars) 5/	658.5	1222.7	1342.5	1484.1	1740.4	1606.1	1171.8	1177.1	1136.6	1308.7	2783.0		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	1.6	1.5	1.3	1.0	1.0	0.9	0.7	0.7	**...**	**1.1**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	48.6	50.0	55.9	48.0	48.9	47.3	41.6	40.6	**...**	**47.8**
Nominal GDP (Million of US dollars)	61,504	67,145	68,538	69,939	75,887	82,900	90,396	98,631	107,756	166,395	373,168		
Nominal dollar GDP growth	1.8	9.2	2.1	2.0	8.5	9.2	9.0	9.1	9.3	8.9	7.8	**8.9**	**8.4**
Memorandum items:													
PV of external debt 7/	...	...	24.7	24.4	22.8	20.6	19.0	17.5	16.2	12.6	9.4		
In percent of exports	...	...	112.4	138.1	129.0	113.2	101.3	93.1	86.3	62.9	41.7		
Total external debt service-to-exports ratio	32.2	27.3	28.5	35.7	33.9	30.4	26.2	24.1	21.7	13.4	6.3		
PV of PPG external debt (in Million of US dollars)			8107.6	8944.4	9704.2	9993.5	10429.1	10752.0	11246.3	15631.0	34957.6		
(PVt-PVt-1)/GDPt-1 (in percent)				1.2	1.1	0.4	0.5	0.4	0.5	0.7	0.8		
Non-interest current account deficit that stabilizes debt ratio	-1.7	2.1	5.1	3.2	4.6	5.6	5.2	5.1	5.1	4.7	5.4		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

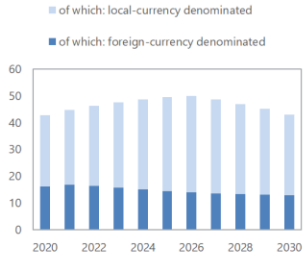
Table 2. Myanmar: Public Sector Debt Sustainability Framework, Baseline Scenario, 2017–40

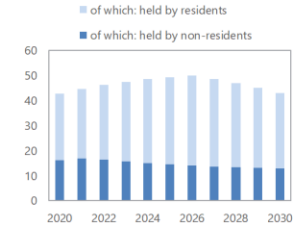
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
Public sector debt 1/	**38.4**	**40.3**	**38.8**	**42.7**	**44.7**	**46.1**	**47.4**	**48.6**	**49.4**	**43.0**	**26.1**	**35.2**	**46.6**
of which: external debt	**15.0**	**16.3**	**14.8**	**16.2**	**16.9**	**16.3**	**15.7**	**15.0**	**14.4**	**12.9**	**12.7**	**13.6**	**14.7**
Change in public sector debt	0.5	1.9	-1.5	3.8	2.0	1.4	1.3	1.1	0.8	-2.0	-1.4		
Identified debt-creating flows	**0.4**	**1.1**	**-0.8**	**3.6**	**1.9**	**1.6**	**1.4**	**1.3**	**1.0**	**-2.0**	**-1.4**	**-0.5**	**0.4**
Primary deficit	**1.5**	**1.6**	**2.5**	**4.2**	**4.1**	**3.4**	**2.9**	**2.6**	**2.3**	**0.3**	**-0.2**	**1.9**	**2.2**
Revenue and grants	17.9	17.6	16.0	14.8	14.7	15.2	16.0	16.3	16.6	17.8	19.9	**17.0**	**16.4**
of which: grants	0.4	0.3	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.3	0.3		
Primary (noninterest) expenditure	19.4	19.2	18.5	19.0	18.8	18.6	18.9	18.9	18.9	18.1	19.7	**18.9**	**18.6**
Automatic debt dynamics	**-0.9**	**-0.4**	**-3.3**	**-0.6**	**-2.1**	**-1.8**	**-1.5**	**-1.3**	**-1.3**	**-2.3**	**-1.2**		
Contribution from interest rate/growth differential	-1.9	-1.8	-2.5	-0.6	-2.1	-1.8	-1.5	-1.3	-1.3	-2.3	-1.2		
of which: contribution from average real interest rate	0.2	0.5	0.0	0.0	0.3	0.9	1.3	1.6	1.7	0.4	0.3		
of which: contribution from real GDP growth	-2.1	-2.3	-2.5	-0.5	-2.4	-2.6	-2.7	-2.8	-3.0	-2.7	-1.4		
Contribution from real exchange rate depreciation	0.9	1.4	-0.8	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**-0.2**	**-0.1**	**0.0**	**-0.1**	**-0.1**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.3**	**0.0**
Privatization receipts (negative)	-0.2	-0.1	0.0	-0.1	-0.1	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**0.1**	**0.7**	**-0.7**	**0.3**	**0.1**	**-0.1**	**-0.1**	**-0.2**	**-0.2**	**-0.1**	**0.0**	**1.3**	**-0.1**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**35.9**	**39.0**	**40.6**	**41.8**	**43.1**	**44.3**	**45.3**	**39.4**	**22.6**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**224.4**	**263.3**	**275.8**	**275.0**	**269.6**	**272.0**	**272.9**	**221.2**	**113.7**		
Debt service-to-revenue and grants ratio 3/	**32.4**	**33.4**	**33.7**	**39.6**	**53.0**	**62.2**	**72.4**	**79.3**	**85.6**	**59.2**	**25.8**		
Gross financing need 4/	7.1	7.4	7.9	10.0	11.8	12.8	14.5	15.5	16.5	10.8	4.9		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	5.8	6.4	6.5	1.4	6.0	6.2	6.3	6.4	6.5	6.4	5.5	**6.6**	**5.9**
Average nominal interest rate on external debt (in percent)	2.3	3.6	2.0	1.1	1.6	1.7	1.6	1.6	1.6	1.9	2.1	**4.4**	**1.6**
Average real interest rate on domestic debt (in percent)	0.1	0.8	-0.2	0.3	1.5	3.4	4.7	5.4	5.5	1.5	1.8	**-1.7**	**3.0**
Real exchange rate depreciation (in percent, + indicates depreciation)	6.2	9.8	-5.3	...	...	...	...	...	...	...	...	**0.9**	**...**
Inflation rate (GDP deflator, in percent)	5.4	5.4	6.3	6.2	7.1	7.0	6.5	6.5	6.5	6.3	6.1	**5.3**	**6.6**
Growth of real primary spending (deflated by GDP deflator, in percent)	-7.3	5.4	2.5	4.1	4.9	5.1	8.1	6.6	6.5	5.7	6.1	**11.9**	**5.7**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	1.0	-0.3	4.0	0.3	2.1	1.9	1.6	1.5	1.5	2.3	1.2	**1.6**	**1.9**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments plus social security, central bank, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Myanmar: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020–30

(In percent)

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of debt-to GDP ratio											
Baseline	13	13	12	12	11	10	10	10	10	10	9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	13	10	7	5	4	2	1	0	-2	-3	-4
B. Bound Tests											
B1. Real GDP growth	13	13	12	12	11	11	10	10	10	10	10
B2. Primary balance	13	13	12	12	11	11	11	10	10	10	10
B3. Exports	13	15	18	17	17	16	15	15	14	14	13
B4. Other flows 3/	13	16	18	18	17	16	15	15	14	14	13
B5. Depreciation	13	16	12	11	11	10	10	10	10	10	10
B6. Combination of B1-B5	13	16	17	16	15	15	14	14	13	13	12
C. Tailored Tests											
C1. Combined contingent liabilities	13	14	13	13	12	12	12	11	11	11	11
C2. Natural disaster	13	14	13	13	12	12	12	12	12	11	11
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	72	72	66	62	58	56	54	52	51	48	47
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	72	57	40	28	19	12	5	-2	-8	-14	-20
B. Bound Tests											
B1. Real GDP growth	72	72	66	62	58	56	54	52	51	48	47
B2. Primary balance	72	73	68	64	61	58	57	56	55	52	50
B3. Exports	72	93	129	120	114	109	106	102	97	89	85
B4. Other flows 3/	72	91	101	94	89	85	83	79	76	69	66
B5. Depreciation	72	72	52	49	46	44	42	41	41	39	39
B6. Combination of B1-B5	72	95	88	96	91	87	84	80	78	72	69
C. Tailored Tests											
C1. Combined contingent liabilities	72	77	72	68	65	63	62	61	60	57	56
C2. Natural disaster	72	79	73	69	66	65	64	63	62	59	58
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	6	6	7	6	6	6	5	5	4	4	4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	6	6	6	5	5	4	3	2	1	0	0
B. Bound Tests											
B1. Real GDP growth	6	6	7	6	6	6	5	5	4	4	4
B2. Primary balance	6	6	7	6	7	6	5	5	4	4	5
B3. Exports	6	7	9	10	10	9	8	9	9	8	8
B4. Other flows 3/	6	6	8	8	8	7	6	7	7	6	6
B5. Depreciation	6	6	7	6	6	5	4	4	3	3	3
B6. Combination of B1-B5	6	7	9	8	9	8	7	7	7	6	7
C. Tailored Tests											
C1. Combined contingent liabilities	6	6	7	7	7	6	5	5	5	5	5
C2. Natural disaster	6	6	7	7	7	6	5	5	5	5	5
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	7	8	8	8	8	6	5	6	5	5	5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	7	8	8	6	6	4	3	2	1	0	0
B. Bound Tests											
B1. Real GDP growth	7	8	9	8	8	7	6	6	5	5	5
B2. Primary balance	7	8	8	8	8	7	6	6	5	5	5
B3. Exports	7	8	9	9	9	8	7	7	7	7	7
B4. Other flows 3/	7	8	9	9	9	8	7	8	8	7	7
B5. Depreciation	7	10	11	9	9	7	6	6	4	5	5
B6. Combination of B1-B5	7	8	10	9	9	8	7	8	7	7	7
C. Tailored Tests											
C1. Combined contingent liabilities	7	8	9	8	8	7	6	6	5	5	5
C2. Natural disaster	7	8	9	8	8	7	6	6	5	5	5
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18
Memorandum item:											
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the threshold.

2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

3/ Includes official and private transfers and FDI.

Table 4. Myanmar: Sensitivity Analysis for Key Indicators of Public Debt 2020–30

(In percent)

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of Debt-to-GDP Ratio											
Baseline	39	41	42	43	44	45	46	45	43	41	39
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	39	39	39	40	41	42	43	43	43	43	43
B. Bound Tests											
B1. Real GDP growth	39	41	44	46	47	49	50	49	48	46	45
B2. Primary balance	39	42	44	46	47	48	48	47	45	43	41
B3. Exports	39	42	48	49	50	51	51	49	47	45	43
B4. Other flows 3/	39	44	48	49	50	51	51	50	48	45	43
B5. Depreciation	39	40	40	39	39	39	38	36	33	30	27
B6. Combination of B1-B5	39	40	41	41	42	43	44	43	41	39	38
C. Tailored Tests											
C1. Combined contingent liabilities	39	47	48	49	50	51	51	49	48	46	44
C2. Natural disaster	39	46	48	49	50	51	52	51	49	47	45
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	263	276	275	270	272	273	272	260	246	235	221
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	263	264	257	249	251	253	257	251	245	244	239
B. Bound Tests											
B1. Real GDP growth	263	281	287	285	290	293	294	284	272	263	251
B2. Primary balance	263	284	291	285	286	286	285	272	257	246	232
B3. Exports	263	289	314	306	306	305	302	288	271	258	242
B4. Other flows 3/	263	298	316	307	307	306	303	289	272	259	242
B5. Depreciation	263	273	261	247	241	234	226	208	189	173	153
B6. Combination of B1-B5	263	272	271	258	261	262	260	249	235	225	211
C. Tailored Tests											
C1. Combined contingent liabilities	263	317	313	305	305	304	302	288	273	261	246
C2. Natural disaster	263	315	313	306	308	309	307	295	281	270	256
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	40	53	62	72	79	86	92	68	65	62	59
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	40	54	60	69	75	81	87	63	62	62	61
B. Bound Tests											
B1. Real GDP growth	40	54	64	76	83	91	97	74	71	69	67
B2. Primary balance	40	53	64	76	83	90	96	72	68	65	62
B3. Exports	40	53	63	74	81	87	93	70	67	65	61
B4. Other flows 3/	40	53	63	74	81	87	93	70	68	65	62
B5. Depreciation	40	50	60	68	74	79	84	62	59	56	52
B6. Combination of B1-B5	40	51	60	70	77	83	89	66	63	60	57
C. Tailored Tests											
C1. Combined contingent liabilities	40	53	72	82	89	95	102	76	72	69	66
C2. Natural disaster	40	54	71	82	89	96	103	77	74	71	68
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.